Amplitude Decay of a Spring–Damper System with Symbolic Coefficients

D. Spring

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Abstract

We write the free vibration of a unit mass on a spring with stiffness k and damping coefficient c as a two-state system. The oscillation is periodic with slowly decaying amplitude, and increasing the damping coefficient increases the decay rate while leaving the period nearly unchanged.

1 The system

In state variables x (displacement) and y (velocity) the dynamics read

$$\dot{x} = y, \tag{1}$$

$$\dot{y} = -kx - cy. \tag{2}$$

2 Initial state and coefficients

The mass is released from rest at unit displacement, with the recorded initial state $x(0) = 1$ and $y(0) = 0$. The nominal coefficients are $k = 2$ and $c = 0.5$.

3 Discussion

The amplitude decays by a fixed factor per cycle set by the damping coefficient, while the oscillation period stays close to the undamped value; small perturbations of the coefficients preserve this picture.